
DSC 140B - Quiz 08

March 12, 2026

Name:

PID:

About the quizzes:

- Quizzes in DSC 140B are *optional* and graded pass/fail.
- A score of 70% or higher earns a “pass” and 1.5 credits toward your final grade.
- If you don’t pass, no credits are earned, but it doesn’t hurt your grade.
- You have 30 minutes to complete the quiz.
- At least one of the questions below will be on an exam (probably with slight changes, such as different numbers).
- Unfortunately, we can’t answer clarifying questions during the quiz. If you think a question has a bug or is unclear, please let us know in a private post on Campuswire after the quiz, and we’ll take it into account when grading.

Problem 1.

A color image of size $64 \times 64 \times 3$ is convolved with a single filter of size $7 \times 7 \times 3$. No padding is applied, and the stride is 1. What is the shape of the output response map?

- ☐ $64 \times 64 \times 3$
- ☐ $58 \times 58 \times 3$
- ☐ $58 \times 58 \times 1$
- ☐ $57 \times 57 \times 1$

Problem 2.

Suppose you have a tabular dataset of 10,000 houses. Each row represents a house, and the columns contain features such as the number of bedrooms, the square footage, the year built, the lot size, and the sale price. You would like to predict the sale price from the other features.

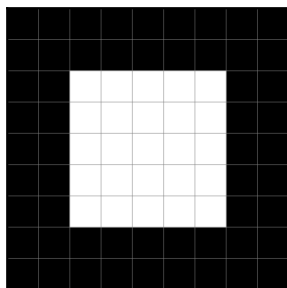
True or False: a convolutional neural network (CNN) would be an appropriate model for this task.

- ☐ True
- ☐ False

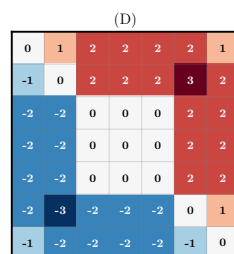
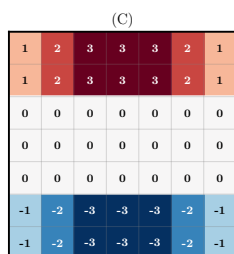
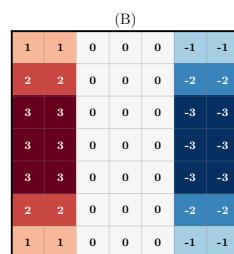
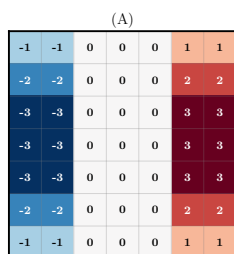
Problem 3.

The image below (where black pixels are 0 and white pixels are 1) is convolved with the following 3×3 filter (stride 1, no padding):

$$\begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$



Which of the following is the resulting response map?



- ☐ (A)
☐ (B)
☐ (C)
☐ (D)

Problem 4.

Consider a convolutional neural network with the following architecture. The input is a $12 \times 12 \times 1$ grayscale image. It passes through Conv layer 1 (4 filters of size 3×3 , stride 1, no padding), producing an output of shape $10 \times 10 \times 4$. Then 2×2 max pooling is applied, producing an output of shape $5 \times 5 \times 4$. Next is Conv layer 2 (6 filters of size $3 \times 3 \times 4$, stride 1, no padding), producing an output of shape $3 \times 3 \times 6$. This is flattened and fed into a fully connected layer with n nodes, followed by an output layer with 1 node.

a) What is the value of n ?

- ☐ 18
- ☐ 36
- ☐ 54
- ☐ 150

b) What is the total number of learnable parameters in the network, excluding biases?

- ☐ 144
- ☐ 252
- ☐ 258
- ☐ 306

Problem 5.

A neural network with 3 output nodes uses the softmax activation function. The pre-activation values (logits) at the output layer are $\vec{z} = (1, 3, 1)$. What is the softmax output $\vec{h} = (h_1, h_2, h_3)$?

- ☐ $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
- ☐ $\left(\frac{e}{2+e^3}, \frac{e^3}{2+e^3}, \frac{e}{2+e^3}\right)$
- ☐ $\left(\frac{e}{2e+e^3}, \frac{e^3}{2e+e^3}, \frac{e}{2e+e^3}\right)$
- ☐ $\left(\frac{1}{2+e^3}, \frac{e^3}{2+e^3}, \frac{1}{2+e^3}\right)$

Problem 6.

A multi-label classifier has 3 output nodes with sigmoid activations. The true labels are $\vec{y} = (0, 1, 1)$ and the predicted probabilities are $\vec{h} = (0.3, 0.7, 0.9)$.

What is the binary cross-entropy loss?

- ☐ $-\log(0.7) - \log(0.7) - \log(0.9)$
- ☐ $-\log(0.3) - \log(0.7) - \log(0.9)$
- ☐ $-\log(0.7) - \log(0.9)$
- ☐ $-\log(0.3) - \log(0.3) - \log(0.1)$

Problem 7.

A multi-class classifier has 4 output nodes with softmax activation. The true label is $\vec{y} = (0, 1, 0, 0)$ and the softmax outputs are $\vec{h} = (0.15, 0.5, 0.25, 0.1)$.

What is the categorical cross-entropy loss?

- ☐ $-\log(0.15) - \log(0.5) - \log(0.25) - \log(0.1)$
- ☐ $-\log(0.85) - \log(0.5) - \log(0.75) - \log(0.9)$
- ☐ $-\log(0.5)$
- ☐ $-\log(0.15) - \log(0.5)$

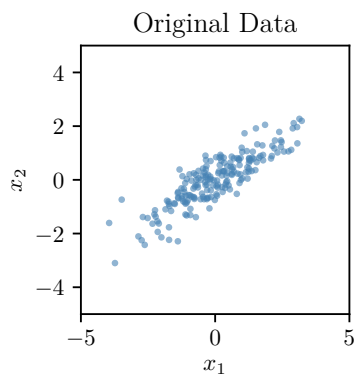
Problem 8.

A neural network classifies images into 5 categories.

- a) Suppose the categories are mutually exclusive (each image belongs to exactly one category). Which pair of output activation function and loss function would be most appropriate?
- ☐ Sigmoid activation with categorical cross-entropy loss
 - ☐ Softmax activation with categorical cross-entropy loss
 - ☐ Softmax activation with binary cross-entropy loss
 - ☐ Sigmoid activation with binary cross-entropy loss
- b) Now suppose an image can belong to multiple categories simultaneously. Which pair of output activation function and loss function would be most appropriate?
- ☐ Sigmoid activation with binary cross-entropy loss
 - ☐ Softmax activation with categorical cross-entropy loss
 - ☐ Sigmoid activation with categorical cross-entropy loss
 - ☐ Softmax activation with binary cross-entropy loss

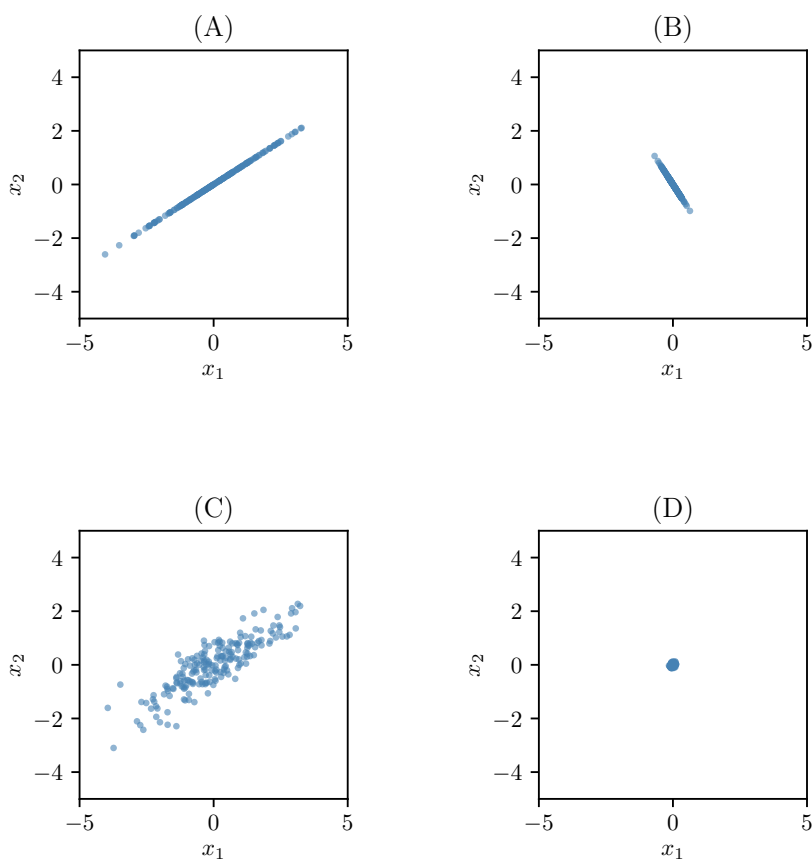
Problem 9.

The plot below shows a dataset of 200 points in $\mathbb{R}^2$.



An autoencoder with 2 input nodes, 1 hidden node, and 2 output nodes (all using linear activations) is trained on this dataset to minimize the mean squared reconstruction error, and the global minimum is achieved.

Which of the following plots shows the decoded data?



- ☐ (A) ☐ (B) ☐ (C) ☐ (D)

Problem 10.

- a) Consider an autoencoder with 10 input nodes, a single hidden layer with 10 nodes, and 10 output nodes. All activation functions are linear (i.e., the identity function).

True or False: given any dataset, the smallest reconstruction error this autoencoder can possibly achieve is zero. You may assume that, when trained, the global minimum of the cost function is achieved.

- ☐ True
☐ False

- b) Now consider an autoencoder with 10 input nodes, 5 hidden nodes, and 10 output nodes. All activation functions are linear (i.e., the identity function), and the network has a single hidden layer.

True or False: there exists a dataset in $\mathbb{R}^{10}$ on which this autoencoder can achieve zero reconstruction error.

- ☐ True
☐ False